

How Many Metastable States Did You Actually Find?

Sampling Budget as a Confound in Dimensionality-Reduction Analyses of Molecular Simulation

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Abstract

A common claim in the molecular simulation literature takes the form “projecting our trajectory with method M reveals N metastable states.” That sentence presents N as a property of the molecule. We show it is also, substantially, a property of how much data was collected. Across two potentials with basin counts fixed by construction and two independent 100 ns all-atom simulations of alanine dipeptide, the number of states reported by PCA, TICA, and VAE projections under standard model-selection criteria increases monotonically with sampling budget while the ground-truth landscape does not change (Spearman $\rho = 0.90$ - 0.97 across systems and methods). The effect survives a coverage-matched control that holds landscape exploration fixed and varies only sample size, which rules out incomplete exploration as the explanation. It is not reducible to a single criterion’s idiosyncrasy: AIC is saturated at every budget tested, BIC inflates monotonically, and only the integrated completed likelihood (ICL) avoids reaching the search ceiling, though it remains inaccurate in a method-dependent direction. Recovery against ground truth under the standard pipeline peaks at an intermediate budget and then declines as more data is added, even as recovery given the true state count remains flat or improves—collecting more data makes the standard analysis pipeline’s answer worse, not better, across every method-system combination tested, including on real molecular dynamics data—with one exception, documented in the text, in which a second simulation seed shifts where the effect begins rather than whether it occurs. We recommend reporting state counts alongside the sampling budget at which they were obtained, and offer ICL as a materially safer default than BIC or AIC for this purpose.

1 Introduction

Dimensionality reduction is the standard first step in characterizing the conformational landscape of a molecular system from simulation data. A trajectory of hundreds or thousands of coordinates is projected onto two or three dimensions using PCA, TICA, or increasingly a learned nonlinear embedding such as a VAE, and the resulting low-dimensional density is read for metastable states—the basins a molecule spends most of its time in and transitions between only rarely.

The sentence that reports this reading, “our analysis reveals N metastable states,” attributes N to the molecule. This paper asks how much of N is instead attributable to the sampling budget: the number of frames the analyst happened to save.

The question matters because sampling budget is rarely fixed by the science. It is set by wall-clock time, hardware access, and project deadlines, and it varies by orders of magnitude across published studies of the same or similar systems. If reported state counts are sensitive to this variable, then a literature comparing state counts across studies—or a single study reporting a state count without its sampling context—is implicitly comparing sampling budgets as much as it is comparing molecules.

We test this directly using systems where the number of metastable states is known independently of any dimensionality reduction or clustering method: two synthetic potentials with basin count fixed by construction, and a real all-atom simulation of alanine dipeptide, whose three conformational states are defined by backbone dihedral angles rather than by any method under audit. On every system tested, the number of states reported by a standard projection-plus-model-selection pipeline increases with sampling budget at fixed ground truth, and a coverage-matched control rules out incomplete exploration as the explanation.

This paper makes three contributions.

A controlled audit of a previously undocumented confound. We show that reported metastable state count is a function of sampling budget under standard analysis pipelines, holding the ground-truth landscape fixed by construction. The effect appears in two synthetic potentials of different dimensionality and basin count, and reproduces in two independent 100 ns all-atom simulations of a real biomolecule.

A control design that isolates the mechanism. By comparing a genuinely short trajectory against a uniform subsample of one long reference trajectory at matched sample size, we separate two candidate explanations that prior comparative work does not distinguish: whether the effect reflects incomplete exploration of the landscape, or the state-count estimator’s response to sample size independent of exploration. The effect persists under coverage matching on every system tested, which locates it in the estimator rather than in what the trajectory visited.

A practical recommendation, not just a diagnosis. Five

model-selection criteria are compared on identical data. BIC and AIC both inflate with budget, with AIC additionally saturated at every budget tested rather than trending; the integrated completed likelihood (ICL) never reaches the search ceiling on either synthetic system, though it is not thereby accurate. We further localize where the resulting error comes from: across all nine method-system pairs tested, recovery under the standard pipeline declines from its peak to the largest budget tested, but the size of the decline is explained by a single mechanism rather than left unexplained. PCA and VAE decline sharply (43-84%) regardless of how faithful their projection is; TICA declines only as much as its warm-up phase—the minimum data needed to produce a stable projection at all—leaves room for the ordinary selection-driven degradation to operate, ranging from 12% where warm-up consumes the entire tested range, through 37% where two budget doublings remain after it, to 53% where most of the range remains.

We do not claim that any published metastable-state count is wrong. The claim is that such counts are, by default, incompletely reported: a number without its sampling budget carries less information than authors and readers typically treat it as carrying.

2 Related work

Comparative benchmarks of dimensionality reduction for molecular systems. Recent work has directly compared PCA, TICA, and VAE projections [1, 2, 3, 4, 5, 6], together with downstream clustering choices, on protein folding trajectories and on Markov state model construction [7, 8, 9], finding that the choice of method changes the resulting free energy surface and the states it appears to reveal; Glielmo et al. [10] survey the field. This literature establishes that method choice matters at fixed data. The present work holds method fixed and varies the data, which is the complementary axis: we ask not which method is best at a given budget, but whether the answer any method gives depends on the budget itself. We are not aware of prior work that varies sampling budget at fixed, independently known ground truth to isolate this effect.

Undersampling and non-ergodicity in molecular dynamics. It is well established and frequently acknowledged that molecular dynamics trajectories are undersampled relative to the timescales of interest, and that this undersampling can bias downstream analysis [11]. This is typically stated as a caveat rather than measured as a quantitative confound with a known ground-truth comparison. Our contribution is to make the undersampling question tractable by constructing systems where the true answer is known, and to show that the resulting effect is not explained by exploration deficit alone, since it persists when landscape coverage is matched across budgets.

Model selection under misspecification. The behavior of BIC and related criteria when the fitted model class does not match the true data-generating process is well studied in the statistics literature [12, 13, 14]; in particular, the likelihood gain from additional mixture components can grow with sample size under misspecification even when the penalty term

does not compensate proportionally. Our results are consistent with this mechanism contributing to the effect, but not with it being the whole explanation: the effect’s persistence under coverage matching, and its variation across a projection method (TICA) whose own stability is separately budget-dependent, point to more than one contributing cause. We are explicit throughout about which part of the effect is attributable to which mechanism, rather than presenting a single unified explanation the data does not fully support.

We are generous to this prior work by design. The gap we identify is narrow: existing benchmarks are good benchmarks, conducted at whatever budget the authors used, and simply do not vary that budget as an independent condition against known ground truth. The contribution of this paper is the control, not a correction to any specific published result.

3 Setup

3.1 Systems

Three systems are used, chosen so that ground truth is available independently of any method under audit.

Müller-Brown potential (2 latent dimensions, 3 basins). A standard three-well potential [15] with minima located by gradient descent from known starting points; basin membership is assigned by steepest-descent quenching to the nearest minimum, never by a fitted clustering model. Figure 1 shows the potential surface, the resulting ground-truth basins, and a representative sampled trajectory.

Prinz potential (1 latent dimension, 4 basins). A standard four-well potential from the Markov state model literature [16]; basin boundaries are the potential’s barrier tops, located once on a fine grid at construction time.

Alanine dipeptide (all-atom, 3 basins). Two independent 100 ns Langevin dynamics simulations in implicit solvent (OpenMM [17], amber14 [18] with GBn2 [19], 2 fs timestep, 1 ps save interval, 100,000 saved frames each). Ground truth is the standard three-region partition of the backbone (ϕ , ψ) dihedral space—a chemical definition entirely independent of dimensionality reduction or clustering. Observed features are pairwise distances between heavy atoms, deliberately excluding ϕ/ψ , so that the methods under audit must recover the landscape from a representation that does not already contain the answer. The simulation length was chosen after an initial 5 ns pilot run showed the rarest basin ($\alpha_L/C7ax$) was visited inconsistently at small sampling budgets; 100 ns ensures all three basins are visited at every tested budget under coverage-matched subsampling, and at budgets of 8,000 frames and above under short trajectories (Section 7).

For the two synthetic potentials, the low-dimensional dynamics are lifted into 30 observed dimensions through a fixed random Fourier feature map plus isotropic noise, drawn once and held constant across every sampling condition, so that no difference between conditions is attributable to a changing observation model. Figure 2 shows the resulting embeddings across projection method and sampling budget on Müller-Brown, coloured by true basin, to illustrate the visual character

Ground truth is known by construction, not inferred

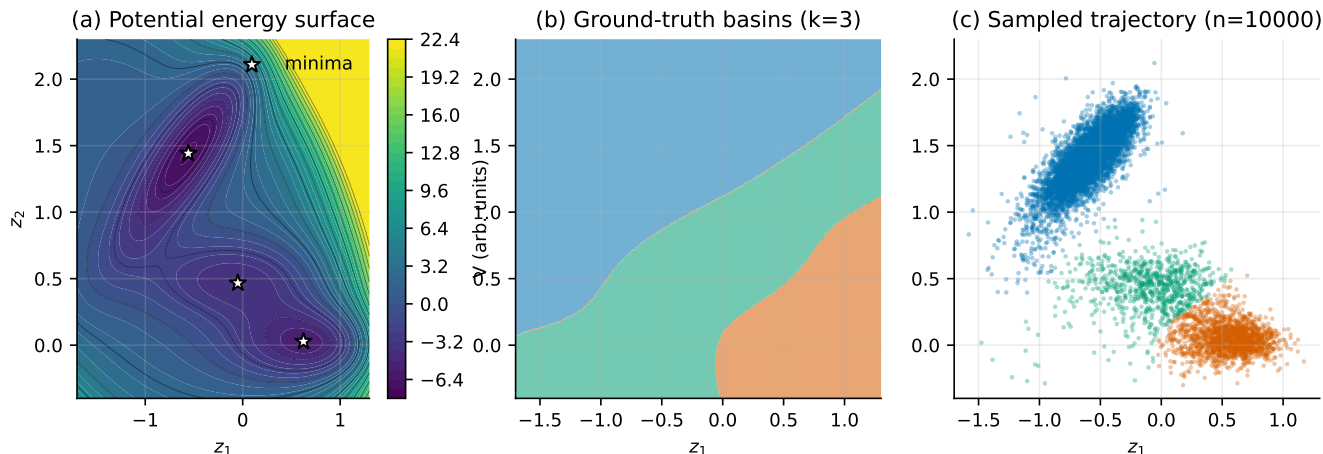


Figure 1: Ground truth for the Müller–Brown potential: (a) the energy surface with its three minima, (b) basin assignment by steepest-descent quenching, (c) one sampled trajectory coloured by true basin. Basin labels never come from a fitted model.

of the confound before the quantitative results in Sections 4-6: the underlying landscape is identical in every panel.

3.2 Sampling conditions

Two modes are compared at each of several sampling budgets (250 to 32,000 frames for the synthetic potentials; 100 to 64,000 frames for alanine dipeptide, whose 100,000-frame trajectory supports a wider sweep than either synthetic system):

- **short:** a trajectory run to produce exactly the target number of saved frames. This is the condition an analyst with a fixed compute budget actually has, and it confounds two things: how much of the landscape was explored, and how many points the selection criterion sees.
- **subsample:** one long reference trajectory per seed, uniformly thinned to the target budget. Landscape coverage is held approximately fixed by construction across every budget; only the number of points varies.

Comparing the two isolates whether an observed effect is due to incomplete exploration (would appear in `short` but not `subsample`) or to the estimator’s response to sample size (would appear in both).

3.3 Methods under audit

PCA [1], **TICA** [2, 3] (solved as an explicit generalized eigenproblem so its regularization is auditable rather than a library default), and a **VAE** [20, 4] (fixed architecture, trained for a fixed number of gradient steps rather than a fixed number of epochs, so that larger datasets are not silently given more optimization). t-SNE [21] is reported only in an appendix figure: its cluster count is governed substantially by its perplexity hyperparameter rather than by data structure [22], so treating its output as a state-count estimate would not be sound.

3.4 State-count selection

Five criteria for choosing the number of mixture components in the projected space are compared on identical embeddings: BIC [12], AIC [23], the integrated completed likelihood (ICL [24], which adds an entropy penalty on component overlap to BIC), silhouette score [25], and an elbow heuristic on k-means inertia. This comparison is included because the most direct objection to the central finding is that it is a known property of BIC’s penalty behavior under model misspecification rather than a property of the dimensionality reduction step; the five-criterion comparison is designed to answer that objection with evidence.

3.5 Metrics and statistics

Recovery is measured by the adjusted Rand index (ARI) [26] between the recovered clustering and ground-truth basin membership, both at the criterion-selected state count and, as a control, at the true state count supplied directly (“oracle k”). ARI is chance-corrected, so it does not reward a method for reporting more clusters—the specific failure mode under investigation. **Basin visitation** and **rarest-basin occupancy** are tracked per condition to identify and exclude degenerate cases in which a trajectory visited fewer true basins than exist, which produce a constant ground-truth label vector against which any single-cluster solution scores $\text{ARI} = 1.0$ trivially.

The seed is the unit of replication: frames within a trajectory are strongly autocorrelated, so all confidence intervals are bootstrapped over seeds, never over frames. The headline statistic reported for the budget effect is a Spearman rank correlation between log sampling budget and reported state count, chosen because the claim under test is monotone inflation rather than any specific functional form. Twenty seeds are used per condition on the synthetic potentials; eight seeds on alanine dipeptide, reflecting its higher per-condition compute cost.

Embeddings coloured by TRUE basin; the landscape is identical in every panel

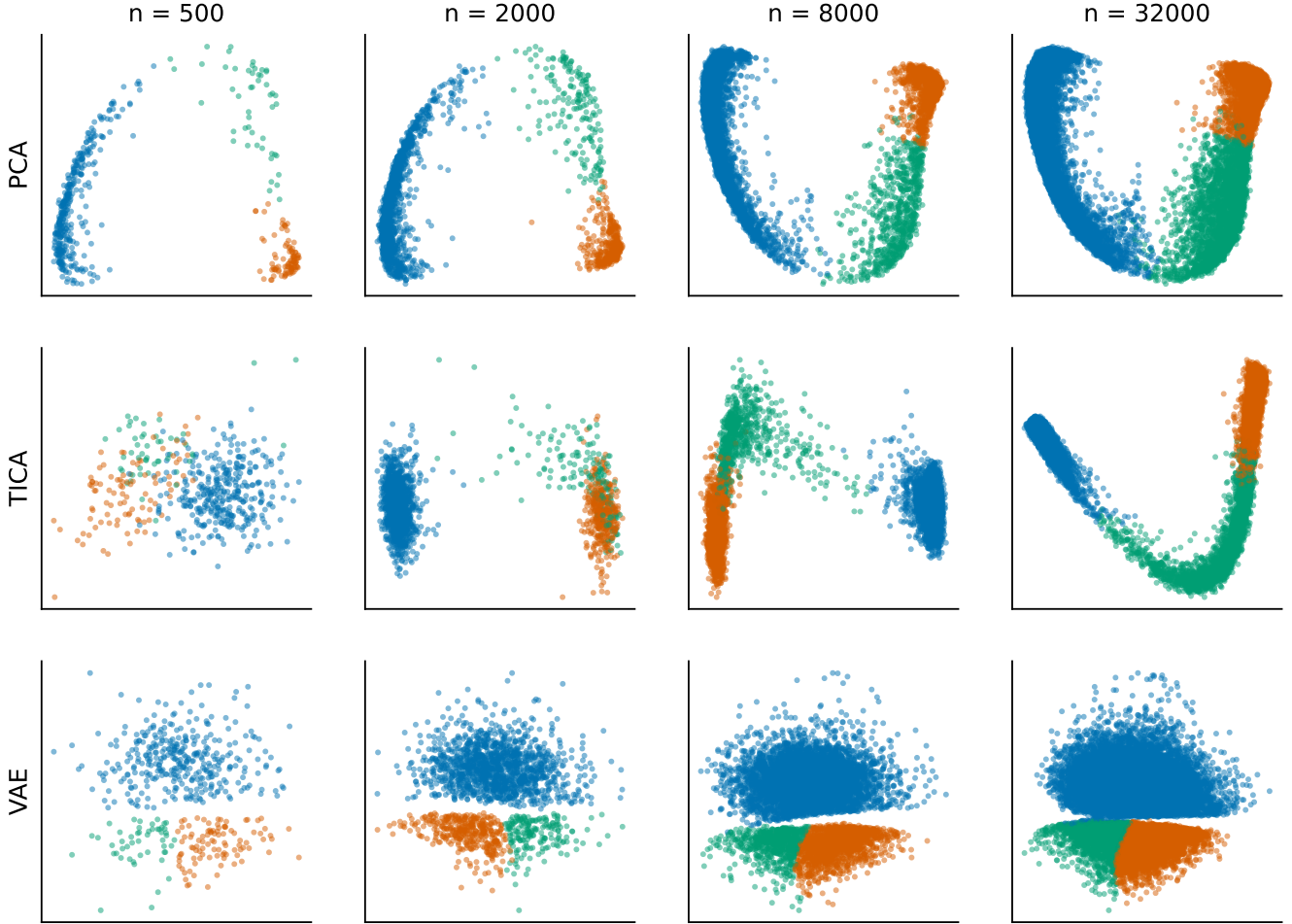


Figure 2: Embeddings of the Müller–Brown system across projection method (rows) and sampling budget (columns), coloured by true basin. The underlying landscape is identical in every panel; only the number of saved frames differs.

4 Reported state count tracks sampling budget

Across both landscapes, all three projections, and both sampling modes, the number of metastable states selected by BIC increases monotonically with the number of saved frames, while the underlying potential is identical in every condition.

On Müller–Brown (true $k = 3$), BIC-selected state counts rise from 1.0–6.3 at $n = 250$ to 11.9–14.8 at $n = 32,000$. On Prinz (true $k = 4$) the same sweep runs from 1.0–7.1 up to 12.5–14.9. At the largest budget every method on every system reports at least three times the true number of basins.

The rank correlation between log sampling budget and BIC-selected state count is uniformly high and tightly bounded (Table 1):

Bootstrap intervals are over the 20 seeds. No interval approaches zero. (Figure 3)

Two points of interpretation are worth making immediately,

because they bound what this result does and does not say.

First, the effect is **not** a claim that these projections fail to represent the landscape. Section 6 shows the opposite. It is a claim about the composite pipeline of projection followed by model selection, which is how state counts are obtained in practice.

Second, the reported counts at large budgets are **right-censored**. The search range was capped at $k = 15$. Under BIC, PCA hits that cap in 44% of Müller–Brown conditions and 29–30% of Prinz conditions (Table 2). Where the cap binds, the tabulated mean is a lower bound on what the criterion would have selected, and the rank correlation is correspondingly attenuated. The reported effect is therefore conservative.

Table 1: Spearman rank correlation between log sampling budget and BIC-selected state count, for each system, sampling mode and projection method. Brackets are bootstrap 95% confidence intervals over the 20 seeds. No interval approaches zero.

system	mode	PCA	TICA	VAE
Müller–Brown	short	+0.88 [+0.85, +0.91]	+0.94 [+0.93, +0.96]	+0.92 [+0.91, +0.94]
Müller–Brown	subsample	+0.88 [+0.85, +0.91]	+0.96 [+0.95, +0.97]	+0.93 [+0.92, +0.94]
Prinz	short	+0.95 [+0.93, +0.96]	+0.97 [+0.97, +0.98]	+0.93 [+0.92, +0.94]
Prinz	subsample	+0.95 [+0.94, +0.96]	+0.96 [+0.95, +0.97]	+0.91 [+0.89, +0.93]

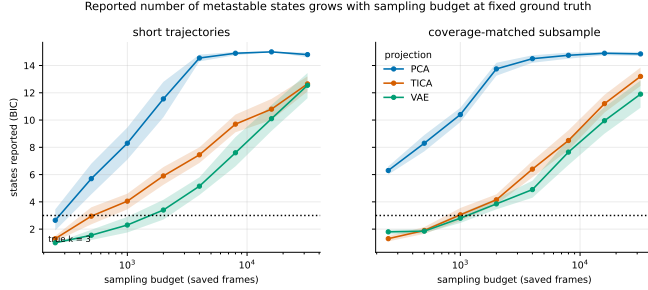


Figure 3: BIC-selected state count against sampling budget, with bootstrap confidence intervals over seeds. The dotted line is the true state count. Counts rise monotonically while the landscape is unchanged.

5 The effect is not reducible to one selection criterion

The obvious objection is that BIC’s penalty grows as $\log n$ while the likelihood gain from an additional mixture component grows with n , so BIC selects more components as n increases under any model misspecification — and a projected free energy basin is never exactly Gaussian. On that account the result would be a property of the criterion rather than of the analysis pipeline.

Five criteria were therefore computed on every condition. The outcome is more informative than a simple yes or no.

AIC is saturated, not budget-driven. AIC reaches the search ceiling in 24–84% of Müller–Brown conditions and 12–65% of Prinz conditions (Table 2) – 65–84% for PCA specifically, 24–28% for TICA and VAE—but Figure 4 shows why that number cannot be read as inflation: AIC’s curve is U-shaped, starting at 14 states for PCA and 8 for TICA and VAE at $n = 250$, dipping near $n = 1000$, then returning to 14.5. It over-selects at every budget, including the smallest. Its rank correlation is computed across a non-monotone curve that spends most of its range censored at the cap and is therefore uninterpretable. AIC is reported for completeness and excluded from the trend claim.

ICL never saturates, but it is not accurate. ICL is the only criterion that reached the ceiling in zero of twelve (system $\times$ mode $\times$ method) cells, which keeps its output in an interpretable range. Within that range it is wrong in a method-dependent direction (Figure 4, Müller–Brown, true $k = 3$): PCA reports 5.0–6.2 states throughout, roughly twice the truth and nearly flat; VAE reports 1.2–1.8, *under-selecting* at every

budget; only TICA rises monotonically through the true value, from 1.0 to 4.6. The defensible claim is that ICL avoids the runaway behaviour of BIC and AIC, not that it recovers the correct state count.

Silhouette and elbow gap are insensitive rather than robust. Both return a near-constant answer regardless of budget. PCA’s silhouette selection is exactly 2 at every budget on Müller–Brown and exactly 4 at every budget on Prinz; elbow gap is flat at 2 for PCA and VAE on both systems. The Prinz case is the more instructive one: silhouette’s constant output there coincides with the true state count, so on that system alone the criterion appears to perform perfectly. It has not measured anything. A criterion that returns the same answer irrespective of the data will look excellent on any landscape where that answer happens to be correct, and this is precisely how an uninformative criterion can acquire a reputation for reliability. The sign changes in Table 3 (silhouette/PCA: -0.26 on Müller–Brown, $+0.65$ on Prinz) are sign changes on an essentially constant output and should be read as noise, not as landscape sensitivity.

The honest summary is narrower than “five criteria disagree” and narrower than a clean ordering by penalty strength: **BIC inflates monotonically with sampling budget; AIC is saturated at all budgets; ICL alone stays in an interpretable range without saturating, though it remains inaccurate in a method-dependent direction; and the two geometry-based criteria are unresponsive to budget and to landscape alike.** (Figure 4)

5.1 The two supporting tables

Ceiling-hit rate—the fraction of conditions in which the criterion selected the largest state count the search allowed ($k_{\max} = 15$). Where this is non-negligible the reported count is right-censored and the inflation is understated (Table 2):

Budget correlation under each criterion, as Spearman ρ between log sampling budget and selected state count. Entries marked “–” are conditions in which the criterion returned the same state count at every budget, so the correlation is undefined—itself the clearest statement of how uninformative silhouette and elbow gap are here:

5.2 One exception, stated plainly

ICL’s resistance does not extend to TICA. TICA’s ICL correlation is $+0.74$ and $+0.92$ on Müller–Brown (short, subsample)

Table 2: Ceiling-hit rate: the fraction of conditions in which the criterion selected the largest state count the search allowed ($k_{\max} = 15$). Where this is non-negligible the reported count is right-censored, so the inflation reported elsewhere is understated. ICL is the only criterion that never reaches the ceiling.

system	mode	criterion	PCA	TICA	VAE
Müller-Brown	short	BIC	44%	1%	4%
Müller-Brown	subsample	BIC	42%	3%	2%
Müller-Brown	short	AIC	66%	27%	24%
Müller-Brown	subsample	AIC	84%	25%	28%
Müller-Brown	short	ICL	0%	0%	0%
Müller-Brown	subsample	ICL	0%	0%	0%
Müller-Brown	short	silhouette	0%	1%	2%
Müller-Brown	subsample	silhouette	0%	1%	0%
Müller-Brown	short	elbow gap	0%	0%	0%
Müller-Brown	subsample	elbow gap	0%	0%	0%
Prinz	short	BIC	29%	0%	9%
Prinz	subsample	BIC	30%	0%	8%
Prinz	short	AIC	62%	19%	37%
Prinz	subsample	AIC	65%	12%	36%
Prinz	short	ICL	0%	0%	0%
Prinz	subsample	ICL	0%	0%	0%
Prinz	short	silhouette	0%	0%	0%
Prinz	subsample	silhouette	0%	1%	0%
Prinz	short	elbow gap	0%	0%	0%
Prinz	subsample	elbow gap	0%	0%	0%

and +0.91 and +0.93 on Prinz — comparable to its BIC correlations rather than to PCA’s and VAE’s ICL values.

We do not have a confirmed mechanism for this. One plausible account is that TICA components, being constructed to maximise autocorrelation rather than variance, produce latent densities less well approximated by a Gaussian mixture, weakening the entropy penalty that ICL relies on. This is a conjecture and is flagged as such; it is a natural target for follow-up rather than something the present data settles. It is reported here because a recommendation to use ICL would be misleading without it.

6 The effect is not explained by incomplete exploration

The competing mechanism is exploration deficit: short trajectories visit less of the landscape, so fewer or differently shaped basins are populated, and the apparent structure changes for reasons that have nothing to do with sample size.

The `subsample` mode addresses this directly. A single long reference trajectory per seed is thinned uniformly to each target budget, so landscape coverage is held approximately fixed by construction while only the number of points varies.

Coverage matching does not remove the effect. BIC budget correlations under `short` and `subsample` are statistically indistinguishable in every cell (Table 4):

Confidence intervals overlap in all six comparisons. Whatever drives the inflation, it survives holding coverage fixed, which locates it in the estimator’s response to sample size rather than in what the trajectory happened to visit. (Figure 5)

6.1 Recovery degrades as sampling budget increases

The gap between recovery at the selected state count and recovery at the true state count is not constant. Recovery under the standard pipeline peaks at an intermediate budget and then declines, while oracle- k recovery over the same range is flat or rising. (Figure 6)

A coverage artifact was checked for before drawing this conclusion. Conditions in which the trajectory visited fewer basins than exist produce a constant ground-truth label vector, against which a single-cluster solution scores $\text{ARI} = 1.0$ trivially. Such conditions occur only in `short` mode at small budgets—100% of Müller-Brown conditions at $n = 250$, declining to zero by $n = 4000$, and 35% of Prinz conditions at $n = 250$, zero by $n = 1000$ —and in `no subsample` condition on either system. All 38 rows scoring $\text{ARI} > 0.999$ are degenerate. Because coverage-matched subsampling contains no degenerate conditions, the primary curves are unaffected: cleaned and raw values agree to two decimal places throughout. Excluding degenerate rows from `short` mode makes the decline steeper rather than shallower (Müller-Brown TICA: peak 0.61 $\rightarrow$ 0.76, final drop -0.42; VAE final drop -0.61), because those rows carried low median ARI (0.023) at the small-budget end. The Müller-Brown `short` condition at $n = 250$ is excluded entirely, having no valid replicates.

Every method-system pair on the two synthetic potentials shows the peak-then-decline pattern under coverage matching; the complete picture across all three systems, including alanine dipeptide, is given as a single table in Section 6.3 rather than duplicated here.

For these pairs, collecting more data makes the answer pro-

Table 3: Budget correlation under each selection criterion: Spearman ρ between log sampling budget and selected state count. Entries marked “–” are conditions in which the criterion returned the same count at every budget, leaving the correlation undefined.

system	mode	criterion	PCA	TICA	VAE
Müller-Brown	short	BIC	+0.88	+0.94	+0.92
Müller-Brown	subsample	BIC	+0.88	+0.96	+0.93
Müller-Brown	short	AIC	+0.61	+0.71	+0.75
Müller-Brown	subsample	AIC	+0.31	+0.67	+0.64
Müller-Brown	short	ICL	+0.58	+0.74	+0.41
Müller-Brown	subsample	ICL	+0.34	+0.92	+0.28
Müller-Brown	short	silhouette	-0.26	+0.14	-0.52
Müller-Brown	subsample	silhouette	–	+0.30	+0.03
Müller-Brown	short	elbow gap	+0.03	+0.37	+0.46
Müller-Brown	subsample	elbow gap	+0.00	+0.54	-0.03
Prinz	short	BIC	+0.95	+0.97	+0.93
Prinz	subsample	BIC	+0.95	+0.96	+0.91
Prinz	short	AIC	+0.67	+0.82	+0.77
Prinz	subsample	AIC	+0.66	+0.85	+0.74
Prinz	short	ICL	+0.48	+0.91	+0.46
Prinz	subsample	ICL	+0.36	+0.93	+0.24
Prinz	short	silhouette	+0.65	+0.26	+0.41
Prinz	subsample	silhouette	+0.12	-0.22	+0.12
Prinz	short	elbow gap	-0.48	+0.28	-0.05
Prinz	subsample	elbow gap	-0.12	+0.09	-0.13

Table 4: The coverage-matched control. BIC budget correlations under genuinely short trajectories and under uniform thinning of one long reference trajectory. Intervals overlap in all six comparisons, so holding landscape coverage fixed does not remove the effect.

system	method	short	subsample
Müller-Brown	PCA	+0.88 [+0.85, +0.91]	+0.88 [+0.85, +0.91]
Müller-Brown	TICA	+0.94 [+0.93, +0.96]	+0.96 [+0.95, +0.97]
Müller-Brown	VAE	+0.92 [+0.91, +0.94]	+0.93 [+0.92, +0.94]
Prinz	PCA	+0.95 [+0.93, +0.96]	+0.95 [+0.94, +0.96]
Prinz	TICA	+0.97 [+0.97, +0.98]	+0.96 [+0.95, +0.97]
Prinz	VAE	+0.93 [+0.92, +0.94]	+0.91 [+0.89, +0.93]

duced by the standard pipeline substantially worse, with the crossover falling between $n = 250$ and $n = 1000$. This is stronger than the endpoint comparison in Table 6, which reports only the largest budget and therefore conceals the non-monotonicity.

6.2 The TICA exception, and the evidence for its mechanism

TICA on Prinz is the exception: recovery rises from 0.00 at $n = 250$ to 0.58 at $n = 8000$ before a small decline to 0.51. It is not an instance of the pattern and is not presented as one.

The mechanism is directly tested, not only inferred from this exception. A lag/lagged-pair ablation (Prinz potential, lags in $\{5, 10, 20, 50, 100\}$, 10 seeds, subsample mode, 300 conditions) fits TICA at every combination of sampling budget and lag and measures oracle-k recovery—the projection’s quality with the selection step removed—as a function of both (Table 5):

Both predictions of the warm-up account are confirmed directly. At any fixed budget, recovery decreases monotonically

as lag increases, consistent with fewer usable lagged pairs ($n_{\text{frames}} - \text{lag}$) producing a less stable time-lagged covariance estimate. The budget at which recovery reaches a given level shifts with lag exactly as the mechanism predicts: 0.35 oracle-k ARI is reached by $n = 1000$ at lag = 5, requires roughly $n = 8000$ at lag = 50, and is not reached even at $n = 16000$ at lag = 100. TICA is therefore subject to two opposing budget effects—an estimator that requires enough lagged pairs to stabilise, with the requirement scaling with the lag itself, and a selection step that degrades with data once the estimator has stabilised (visible in the corresponding selected-k table, where recovery at lag = 5 peaks at $n = 8000$, 0.56, and declines to 0.45 by $n = 16000$, reproducing the main peak-then-decline pattern within a single fixed lag setting)—and the observed curve on any given system is their sum. Where the crossover falls depends on how quickly TICA stabilises on that landscape’s combination of lag and dynamics: near $n = 1000$ on Müller-Brown at the default lag, near $n = 8000$ on Prinz. PCA and VAE have no comparable warm-up requirement, consistent with their declining monotonically or peaking at the smallest budgets in the main sweep.

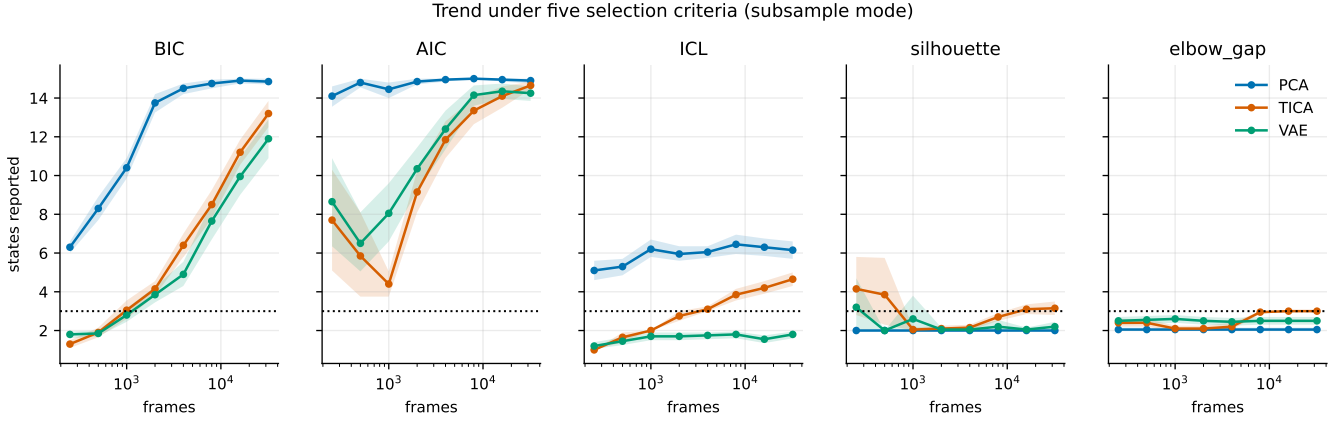


Figure 4: The same trend under all five selection criteria. BIC inflates monotonically; AIC is saturated at every budget rather than trending; ICL alone stays in an interpretable range; silhouette and elbow gap are insensitive to budget and to landscape alike.

Table 5: TICA lag ablation on the Prinz potential: oracle- k recovery as a function of sampling budget and lag (10 seeds, subsample mode, 300 conditions). Recovery falls monotonically as lag grows at fixed budget, and the budget needed to reach any given level rises with lag.

n_frames	lag=5	lag=10	lag=20	lag=50	lag=100
500	0.10	0.03	0.03	0.05	0.03
1000	0.35	0.16	0.06	0.06	0.07
2000	0.53	0.40	0.24	0.05	0.06
4000	0.67	0.55	0.42	0.20	0.09
8000	0.77	0.69	0.59	0.44	0.29
16000	0.87	0.80	0.70	0.57	0.54

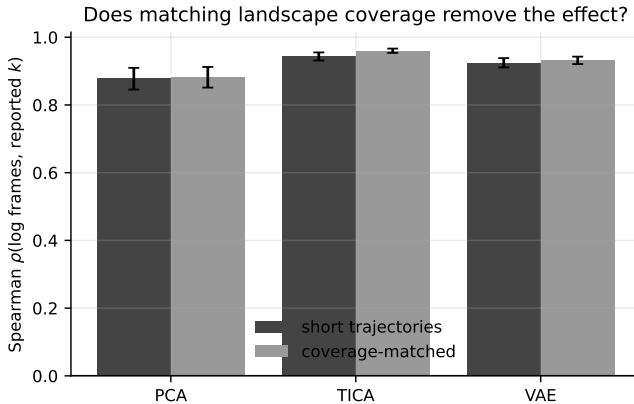


Figure 5: Budget correlation with and without coverage matching. The effect survives holding landscape exploration fixed, which places it in the estimator rather than in what the trajectory visited.

6.3 Scope of the localisation claim

Oracle- k recovery separates cleanly by system rather than by method. On Prinz, all three projections reach 0.81-0.90 when given the true state count, so the projections are faithful and essentially all of the observed degradation is attributable to state-count selection. On Müller-Brown only TICA reaches that range (0.85); PCA and VAE reach only 0.41-0.44 even with the true k supplied, so for those pairs the projection is itself a limiting factor and selection is not the whole story.

The claim that the projection is faithful and only the selection step fails is therefore supported wherever oracle- k recovery is high—all of Prinz, and TICA on Müller-Brown—and is not supported in general. The paper should scope it accordingly rather than asserting it as a universal result.

(All values at $n = 32,000$, short mode; subsample values are within 0.03. Note that these endpoint figures understate oracle- k recovery for TICA under subsampling, which reaches 0.82 on Prinz at the largest budget.)

6.4 The complete picture across nine method-system pairs

Every method-system pair tested shows recovery declining from its peak to the largest budget tested. What differs is how much, and the amount tracks a specific mechanism rather than varying arbitrarily.

Two patterns are visible in this table together that were not visible when the systems were reported separately.

PCA and VAE decline sharply and consistently (43-84% relative decline) across all three systems, regardless of how faithful the underlying projection is (compare their oracle- k column, which ranges from 0.42 to 0.90). Their selected- k recovery is dominated by the state-count selection failure, not by projection quality.

TICA’s decline tracks how much budget remains after its warm-up, not the system. TICA requires a minimum number of usable lagged pairs to produce a stable projection

Table 6: Recovery at the criterion-selected state count against recovery given the true state count, at $n = 32,000$ in `short` mode. The gap localises how much of the error belongs to the selection step rather than to the projection.

system	method	selected k	oracle k	gap
Müller-Brown	PCA	0.102	0.415	0.31
Müller-Brown	TICA	0.342	0.847	0.51
Müller-Brown	VAE	0.111	0.423	0.31
Prinz	PCA	0.349	0.897	0.55
Prinz	TICA	0.482	0.815	0.33
Prinz	VAE	0.399	0.854	0.46

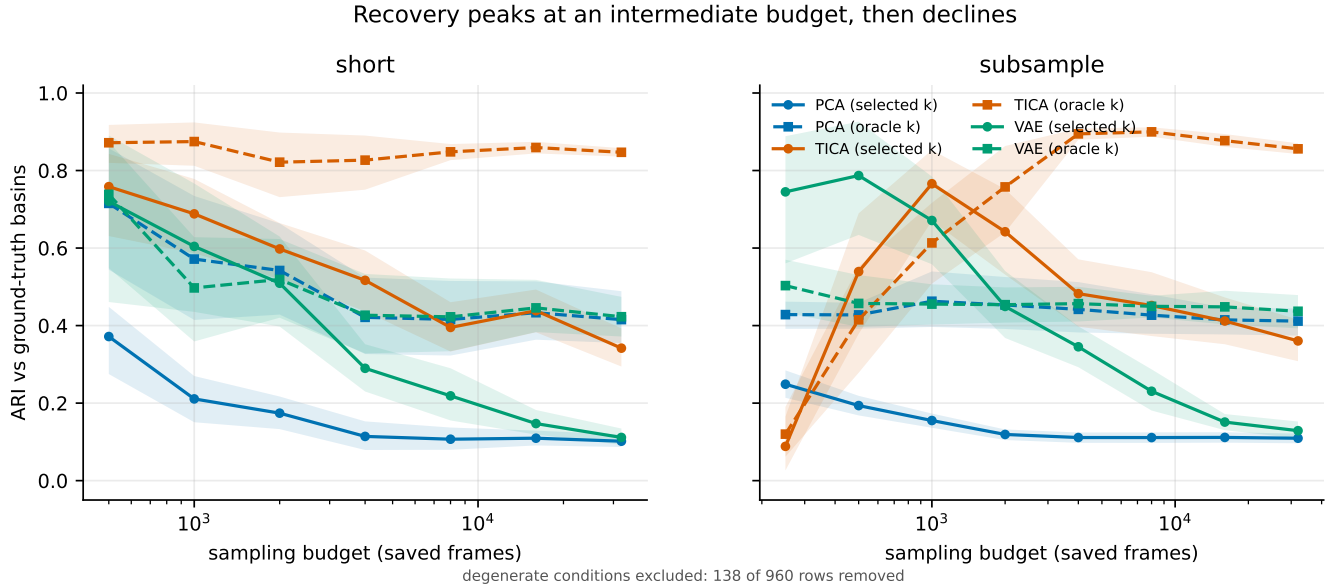


Figure 6: Recovery at the selected state count against recovery given the true state count. Selected- k recovery peaks at an intermediate budget and then declines while oracle- k recovery is flat or rising: more data makes the standard pipeline’s answer worse. Degenerate conditions, in which fewer basins were visited than exist, are excluded.

at all; at small budgets on every system it starts near zero (Müller-Brown: 0.09 at $n=250$; Prinz: 0.00 at $n=250-500$; alanine: 0.00 at $n=100$). The right diagnostic for when warm-up has finished is the oracle- k column, which measures projection quality with the selection step removed. Reading warm-up off that column, the size of the subsequent decline is a simple function of how many budget doublings the sweep has left once warm-up completes (Table 8):

Prinz spends its entire range warming up and barely declines; Müller-Brown finishes early and declines as sharply as PCA and VAE on the same system; alanine dipeptide falls between them because its warm-up ends two doublings before the sweep does. This is two mechanisms operating on one curve rather than one: an estimator-stability effect that dominates at small budgets, and a selection-driven degradation that dominates once the estimator has stabilized, exactly as characterized via the oracle- k contrast in Section 6.

Why alanine’s warm-up completes so late, and a caution it implies. That alanine warms up at $n=16,000$ is at first surprising, since its reference trajectory is the longest in the paper. The explanation is a direct consequence of the lag mechanism

established in Section 6, and it is specific to the `subsample` condition. TICA’s lag is specified in frames of the data it is handed, but uniform thinning to a budget of n frames from a 100,000-frame trajectory imposes a stride of $100,000/n$, so a nominal lag of 10 frames corresponds to an *effective* physical lag of $10 \times 100,000/n$ picoseconds. The effective lag therefore shrinks as the budget grows, and TICA’s projection quality tracks it closely (Table 9):

Oracle- k recovery is flat at roughly 0.25 across six consecutive budgets and then rises nearly fourfold between $n=8,000$ and $n=16,000$ —precisely where the effective lag crosses below about 100 ps, the timescale of the backbone dihedral transitions that separate these basins. Above that lag the time-lagged covariance is estimated between frames that are already decorrelated, and TICA has no slow direction left to find. The same figures in `short` mode, where the stride is always 1 and the effective lag is fixed at 10 ps, show no such discontinuity: oracle- k rises smoothly from 0.15 to 0.96. Figure 7 plots both modes against budget and against effective lag; the discontinuity is present in the first view and resolved by the second.

The practical caution follows directly, and it is not one we

Table 7: All nine method–system pairs. Every pair declines from its peak to the largest budget tested; the size of the decline tracks how much budget remains after TICA’s warm-up completes, and is otherwise uniformly large for PCA and VAE.

system	method	peak ARI (budget)	ARI at largest budget	relative decline	oracle-k ARI
Müller-Brown	PCA	0.25 (n=250)	0.11	56%	0.415
Müller-Brown	TICA	0.77 (n=1000)	0.36	53%	0.847
Müller-Brown	VAE	0.79 (n=500)	0.13	84%	0.423
Prinz	PCA	0.64 (n=250)	0.35	45%	0.897
Prinz	TICA	0.58 (n=8000)	0.51	12%	0.815
Prinz	VAE	0.72 (n=1000)	0.41	43%	0.854
Alanine dipeptide	PCA	0.47 (n=100)	0.12	74%	0.475
Alanine dipeptide	TICA	0.44 (n=16000)	0.27	37%	0.963
Alanine dipeptide	VAE	0.54 (n=2000)	0.18	66%	0.421

Table 8: Why TICA’s decline varies by system. Warm-up completion is read from the oracle-*k* column; what remains of the sweep after it determines how much ordinary selection-driven degradation can occur.

system	warm-up completes at	doublings remaining in sweep	decline
Prinz	not before n=32,000 (end of range)	0	12%
Alanine dipeptide	n=16,000	2	37%
Müller-Brown	n≈1,000-4,000	3-5	53%

set out to test: thinning a trajectory before applying TICA rescales the lag along with the data. An analyst who subsamples aggressively and keeps the lag parameter fixed is silently lengthening the effective lag, and can push a well-sampled trajectory back into the warm-up regime. This is an independent confirmation of the lag ablation of Section 6 on real molecular dynamics, arrived at through the thinning stride rather than through the lag parameter itself.

All nine method-system pairs decline from peak to the largest budget tested, and the size of the decline is explained by a specific, testable mechanism rather than left as an unexplained residual in a minority of cases. This table, not a summary fraction, is the result that belongs in the paper.

One qualification, established by the second alanine seed reported in Section 7 and not visible in this table. The decline column for TICA depends on where that system’s warm-up happens to complete, and the completion budget is itself seed-dependent: on a second independent 100 ns trajectory TICA’s warm-up finishes one doubling later, at n=32,000, and TICA consequently shows no decline at all within the tested range. The mechanism predicts this—zero doublings remaining, as on Prinz—but it means the TICA rows here should be read as one draw from a seed-dependent quantity rather than as a fixed property of the system. The PCA and VAE rows replicate closely across seeds.

7 Alanine dipeptide: simulation detail and comparison to the synthetic systems

The synthetic systems establish the effect under conditions where ground truth is exact and the observation model is fully controlled. This section reports the real-molecule replication

and what is and is not comparable between the two settings.

Simulation. A single 100 ns trajectory of alanine dipeptide (ACE-ALA-NME) was run in implicit solvent (OpenMM [17], amber14 [18] with the GBn2 implicit solvent model [19], Langevin middle integrator [27], 2 fs timestep, 300 K, 1 ps save interval), producing 100,000 saved frames. A second trajectory was run identically from a different random seed; unless stated otherwise the results below are from the first, and the two are compared at the end of this section. Dihedrals and heavy-atom distances were computed with MDTraj [28]. An initial 5 ns pilot run was discarded after it showed the rarest of the three basins (alpha_L/C7ax) was not consistently visited at the sampling budgets under study; 100 ns was chosen because it visits all three basins at every tested budget under coverage-matched subsampling, and at budgets of 8,000 frames and above under short trajectories (Table 10). Sampling budgets from 100 to 64,000 frames were drawn from this trajectory: `short` budgets as contiguous leading blocks, `subsample` budgets as uniform thinning, matching the two conditions used on the synthetic systems. Ground-truth basin membership was assigned from the backbone (phi, psi) dihedral angles into the standard three-region partition (C7eq, alpha_R, alpha_L/C7ax), a chemical definition independent of any method under audit. Figure 8 shows, for the seed-1 trajectory, that partition and the resulting basin populations: occupancy is strongly uneven, at 0.639, 0.308 and 0.053, and it is the 5.3% basin that makes the sampling budget bite. At n = 100 frames a basin that rare is expected to contribute about five frames, which is why the 5 ns pilot reached it inconsistently and why coverage matching matters so much on this system. Observed features for the dimensionality reduction methods were pairwise distances between heavy atoms, excluding phi/psi so that the audited methods must recover the landscape from a representation that does not already encode

Table 9: TICA under coverage-matched subsampling on alanine dipeptide. Thinning to n frames from a 100,000-frame reference imposes a stride of $100,000/n$, so the effective physical lag shrinks as the budget grows. Projection quality tracks the effective lag, not the budget.

n_frames	100	1,000	8,000	16,000	64,000
effective lag	10 ns	1 ns	125 ps	62 ps	16 ps
oracle-k ARI	0.02	0.26	0.25	0.93	0.96

Thinning a trajectory rescales TICA's lag: warm-up tracks τ_{eff} , not sample size

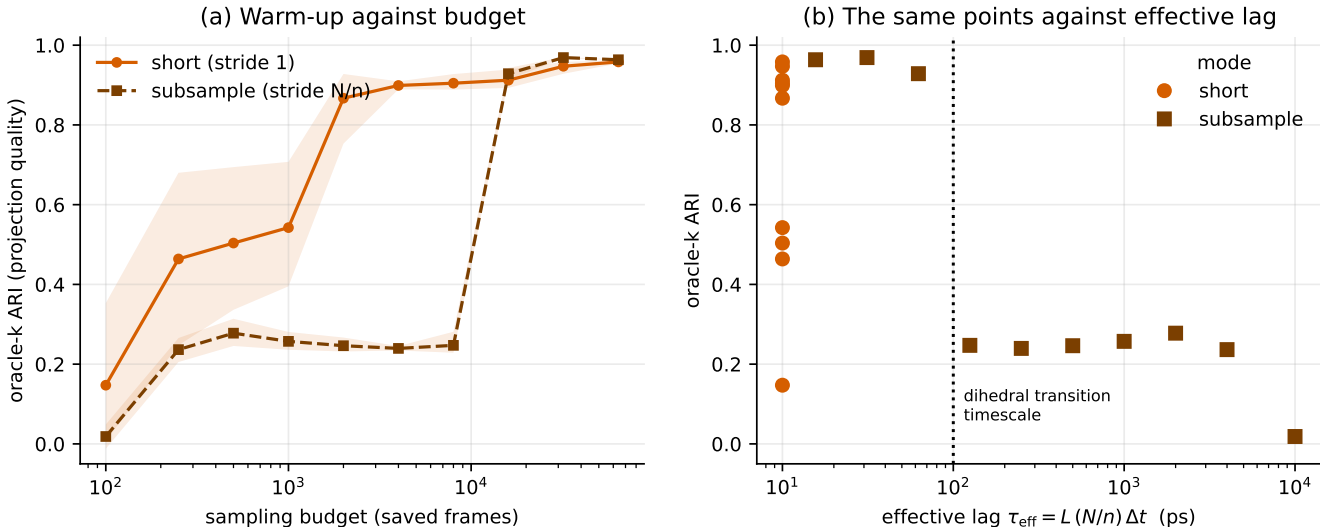


Figure 7: Thinning a trajectory rescales TICA’s lag. (a) Warm-up against sampling budget for both modes. (b) The same points against the effective lag $\tau_{\text{eff}} = L(N/n)\Delta t$: every subsample point above the dihedral transition timescale sits near 0.25 and everything below it jumps to roughly 0.95, so lag alone accounts for that curve. The *short* points sit at a fixed 10 ps and still span 0.15 to 0.96, so their spread is sample size alone.

the answer.

Under coverage-matched subsampling, all three basins are visited at every budget tested, from $n=100$ upward, since subsampling draws uniformly from the full 100,000-frame reference trajectory rather than from a contiguous block.

Replication of the main effect. BIC-selected state count increases with sampling budget on all three methods, in both modes (Spearman $\rho = 0.90$ - 0.97 , 8 seeds per correlation—fewer than the 20 seeds used on the synthetic systems, per the limitation discussed in Section 9). PCA rises from a mean of 3.1 states at $n=100$ to 14.5 at $n=64,000$, against a true count of 3. TICA and VAE rise from conservative counts near 1 at the smallest budget to 13.9 at the largest. All three methods approach the search ceiling ($k_{\text{max}}=15$) at the largest budgets tested, consistent with the ceiling-censoring pattern documented on the synthetic systems.

A method ranking that reverses relative to the synthetic systems. On the synthetic potentials, TICA is not uniformly the best-recovering method (it matches PCA and VAE’s oracle-k quality only on Müller-Brown and is comparable to all three on Prinz). On alanine dipeptide, TICA’s oracle-k recovery (0.958) is clearly the best of the three, well above PCA (0.494) and VAE (0.391) – the largest gap between TICA and the other two methods observed on any system in this pa-

per. Its selected-k recovery at the largest budget (0.249) remains far below this ceiling, the same selection-step bottleneck seen throughout. This is consistent with TICA’s construction: it is designed to find slow collective coordinates, and backbone dihedral transitions in a real molecule are exactly the kind of slow process TICA targets, whereas the Müller-Brown and Prinz potentials were not constructed to favor any particular method. The practical implication is that TICA’s relative advantage may be larger on real molecular systems than the synthetic benchmarks alone would suggest—a claim the synthetic systems could not have supported on their own, and that alanine dipeptide is well-suited to test given how it was built.

Replication on a second independent 100 ns seed. A second 100 ns trajectory was run under an identical protocol from a different random seed and swept over the budgets the two runs share (100 to 32,000 frames). This is a genuine physical replicate: it reran the full Langevin sampling process, not merely the subsampling and clustering randomness that the eight sweep seeds vary (Table 11).

The central claim replicates closely. No budget-inflation correlation moves by more than 0.02 between seeds, the method ranking is unchanged with TICA’s oracle-k far above PCA’s and VAE’s on both, and basin visitation behaves identically (all three basins reached by $n=8,000$ under short trajec-

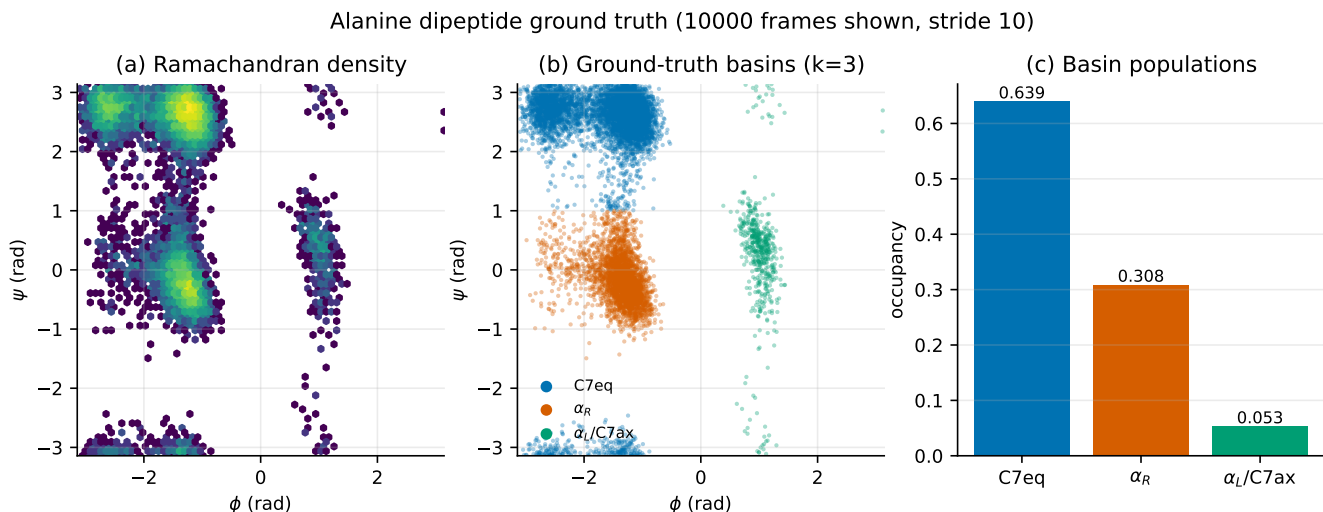


Figure 8: Alanine dipeptide ground truth, from the seed-1 trajectory: (a) Ramachandran density, (b) the standard three-region partition that supplies the basin labels, (c) basin occupancy. The partition is a chemical definition and is independent of every method under audit. The rarest basin holds 5.3% of frames, which is why sampling budget bites hardest on this system.

Table 10: Mean number of the three ground-truth basins visited by contiguous leading blocks of the 100 ns trajectory. Under coverage-matched subsampling all three are visited at every budget.

n_frames (short mode)	100	250	500	1000	2000	4000	8000+
mean basins visited (of 3)	2.0	2.1	2.1	2.3	2.6	2.8	3.0

ories and at every budget under subsampling).

One quantity does not replicate, and it is worth stating plainly rather than averaging away: **TICA shows no decline at all on seed 1 within the tested range.** Its recovery rises monotonically to the largest budget, where seed 0 had peaked at $n=16,000$ and fallen 27% by $n=32,000$. This is not a contradiction of the mechanism but a consequence of it. Section 6.3 argues that TICA’s decline is set by how many budget doublings remain after its warm-up completes; on seed 1 the warm-up completes one doubling later, at $n=32,000$ rather than $n=16,000$, leaving zero doublings in which selection-driven degradation could act. The oracle- k column shows the shift directly: at $n=16,000$ seed 0 has already reached 0.93 while seed 1 is only midway at 0.52, and both arrive at 0.96 by $n=32,000$. The seed-1 alanine case is therefore the same configuration as Prinz—warm-up consuming the entire sweep, zero doublings remaining, no decline observed—reached on a different system.

What this costs the paper is precision about *where* TICA’s crossover falls, not the existence of the effect: the crossover budget is itself seed-dependent, so a single seed cannot pin it down. What it buys is a prediction confirmed out of sample, since the doublings-remaining account was formulated on seed 0 and correctly anticipates a null result on seed 1.

8 Recommendations

1. **Report the sampling budget alongside any claimed metastable state count.** A state count without its budget is, by the evidence in this paper, an incompletely specified quantity.
2. **Do not report a state count from a single budget as a stable property of the system.** Where feasible, report how the count changes with budget (as in Table 7), or explicitly justify why the budget used is believed to be past the point where this matters—which, per point 3, requires checking rather than assuming.
3. **Check the gap between recovery at the selected state count and at a plausible alternative before trusting the selected count, rather than assuming the selection step is reliable.** This paper’s oracle- k comparison is one way to do this when ground truth is known; in practice a domain plausible-range check (does $k = 3$ versus $k = 8$ tell qualitatively different biological stories?) serves a similar function.
4. **Prefer ICL over BIC or AIC for this purpose, with the caveat that ICL is not thereby accurate, only less prone to runaway inflation.** ICL never reached the search ceiling in any of the twelve (system $\times$ mode $\times$ method) conditions tested here, where BIC did so in up to 44% of conditions and AIC in up to 84%. ICL’s own

Table 11: Two independent 100 ns alanine dipeptide seeds, compared over the budgets they share. Budget inflation, the method ranking and basin coverage all replicate. TICA’s relative decline does not: its warm-up completes one budget doubling later on seed 1, leaving no room for selection-driven degradation within the tested range, exactly as the doublings-remaining account predicts.

quantity (budgets 100-32,000)	seed 0	seed 1
budget-inflation ρ , range over 6 mode x method cells	+0.87 to +0.97	+0.88 to +0.97
PCA relative decline, subsample	73%	73%
VAE relative decline, subsample	66%	63%
TICA relative decline, subsample	27%	0%
PCA oracle-k at n=32,000	0.507	0.546
TICA oracle-k at n=32,000	0.947	0.903
VAE oracle-k at n=32,000	0.466	0.524
TICA warm-up completes at	n=16,000	n=32,000

reported count should still be treated as an estimate to sanity-check, not a final answer.

5. **Where TICA is used, check for evidence of a warm-up regime before trusting its output at a given budget.** TICA’s failure mode at small budgets (near-zero recovery, not merely inaccurate recovery) is distinct from PCA’s and VAE’s and is not fixed by any of the five selection criteria tested; it requires enough usable lagged pairs (sampling budget minus lag) at the chosen lag to produce a stable estimate. This is not only a plausible diagnostic but a measured one: the lag ablation in Section 6 shows recovery at fixed budget falling monotonically as lag increases, and the budget needed to reach any given recovery level rising correspondingly with lag. A practical check even without ground truth is whether the projection changes substantially under a moderate change in lag; if it does, the estimate at the current lag and budget should not yet be trusted.

9 Limitations

Ordered by how much they constrain the paper’s claims, most binding first.

TICA’s warm-up completion budget is seed-dependent, so the size of its decline is not a fixed property of a system. The two 100 ns alanine seeds place that crossover one budget doubling apart, which is the difference between a 27% decline and none at all within the tested range (Section 7). The doublings-remaining mechanism accounts for both, and PCA and VAE replicate closely, but any single-seed estimate of *how far* TICA declines on a given system should be treated as one draw rather than a constant. Pinning the crossover down would need several seeds per system, which is not done here.

Two independent 100 ns seeds on alanine dipeptide, against twenty on each synthetic system. The real-system evidence is now genuinely replicated rather than single-seed, but two is still far short of the twenty independent Langevin runs behind each synthetic result, and the eight sweep seeds within each trajectory vary only subsampling, clustering initialization and VAE training rather than the underlying dynamics. Quantities that agreed closely across the two seeds are re-

ported as replicated; the one that did not is reported as such in Section 7 rather than averaged.

An earlier 5 ns cross-seed comparison is retained only as provenance. Before the 100 ns runs, two independent 5 ns simulations were swept and compared, and the budget-inflation effect replicated across them. Both shared the basin-visitation deficit that motivated the move to 100 ns, so that agreement was never strong evidence about the properly-sampled result. It is now superseded outright by the 100 ns cross-seed comparison in Section 7 and is mentioned here only so the discarded pilot data in the repository is accounted for.

Silhouette and elbow-gap criteria are shown to be unreliable but not fully explained. Both return near-constant output regardless of budget, and this paper documents that fact and its consequence (an illusion of stability that happens to coincide with the correct answer on some systems and not others) without establishing why these particular criteria are so insensitive to this particular kind of data. This is noted as an open question rather than resolved.

All three systems tested have a small number of well-separated basins (3-4). Whether the effect’s magnitude changes for landscapes with many close or overlapping metastable states is untested and is a natural extension.

Alanine dipeptide, while a real molecule, is small and fast-mixing. The result should not be extrapolated to slower-folding or larger proteins without further validation; the paper’s claims about real systems are scoped to this system and are not evidence about protein folding timescales generally.

Two selection-criterion claims rest on limited excursions. ICL’s zero ceiling-hit rate is a strong result but was tested only up to a search ceiling of $k_{\max} = 15$; whether ICL would eventually saturate at a still-larger budget is not established. Separately, ICL’s own accuracy (as opposed to its resistance to runaway inflation) has not been separately validated against an independent ground truth beyond the ARI figures already reported.

TICA’s lag parameter was varied on one system only. The warm-up mechanism proposed in Section 6 is now directly tested via a lag/lagged-pair ablation (300 conditions, 10 seeds) rather than only inferred from the oracle-k contrast, and the ablation confirms both qualitative predictions of the mechanism. The lag parameter itself was swept on the Prinz po-

tential only. Section 6.3 supplies a second, independent test on alanine dipeptide by varying the effective lag through the subsampling stride rather than through the parameter, and the crossover falls where the mechanism predicts; a direct parameter sweep on Müller-Brown and on alanine remains untested and would further strengthen the claim’s generality.

The embedding-dimension ablation is complete and confirms the main results are dimension-robust, with two secondary findings worth reporting. Re-running the core measurement at `n_components` in `{2, 3, 5}` on Prinz (540 conditions, 10 seeds) shows the budget-inflation effect essentially unchanged across dimension for all three methods (Spearman ρ stays within 0.86-0.98 with no consistent trend as dimension increases), and ICL’s zero ceiling-hit rate holds at all three dimensions (0 of 9 method-by-dimension combinations reach the cap), strengthening Recommendation 4 beyond the single dimension used elsewhere in the paper.

Two secondary effects emerged that were not anticipated. PCA’s oracle-k recovery improves monotonically with dimension (0.895 to 0.958 from `dim = 2` to `dim = 5` at the largest budget), consistent with a purely linear projection retaining more information as more output dimensions are allowed; TICA’s and VAE’s oracle-k recovery is roughly flat or mildly declines over the same range, consistent with both already capturing the dynamically relevant directions at two dimensions, leaving additional dimensions to add unneeded capacity (VAE) or extra room for the selection step to overfit (TICA). Because selected-k recovery barely moves with dimension for any method, PCA’s oracle-versus- selected gap widens as dimension increases—a cleaner illustration of the paper’s localization claim than the two-dimensional case alone, since it shows the selection step’s relative cost growing precisely as the projection itself improves. This ablation was run on Prinz only; whether the same pattern holds on Müller-Brown and alanine dipeptide is untested.

Data availability

All result files underlying every number and figure in this paper are in the accompanying repository at <https://github.com/imjbassi/sampling-audit>, under `results/`: `sweep.csv` and `sweep_prinz.csv` (960 conditions each), `alanine_sweep_100ns.csv` (480 conditions), `tica_lag_ablation.csv` (300 conditions) and `dim_ablation.csv` (540 conditions). The superseded 5 ns alanine pilot sweeps are retained in the same directory for provenance and are identified as such in `results/NOTE.md`. Raw trajectories are not distributed because they are large and exactly reproducible from the simulation script; the alanine dipeptide starting structure is included.

Code availability

All code is in the same repository under the MIT License. `run_all.sh` reproduces the synthetic sweeps

end to end; `src/alanine.py` runs the molecular dynamics and its sweep; `src/figures.py` and `src/figures_alanine.py` regenerate every figure from the result CSVs.

Competing interests

The author declares no competing interests.

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